Pitch:
Learning and generating structured graphs is key to applications such as molecular design, drug discovery, and evolving relational systems. Existing approaches generally model graph generation as a sequential decision process or through temporal embeddings. These methods usually lack an explicit, interpretable model of the dynamics of how the graphs evolve, and they lack the ability to simulate future states and support planning.

This work investigates whether structured dynamics models can learn and predict graph evolution in latent space more effectively than standard neural transition models.

Hypothesis: A linear operator in a latent space can provide more stable and interpretable multi-step predictions of graph evolution than nonlinear neural transition models by capturing dynamics in a lifted representation where they may be approximated linearly. While Koopman operator theory is classical for continuous dynamical systems, recent work on latent space linearization, such as Lusch et al., suggests that expressive encoders can map complex dynamics into spaces where linear evolution becomes tractable. Applying this intuition to graph domains is non-trivial, given their discrete and combinatorial nature, making this comparison empirically interesting.

RL-based approaches, such as GCPN, formulate graph generation as a sequential decision process (MDP), but rely on environment-defined transitions without learning explicit dynamics. Temporal graph models such as TGN capture evolution via event-driven embedding updates, but they do not model state transitions explicitly. In contrast, model-based RL in other domains learns transition dynamics that enable planning. This suggests a gap in explicit graph dynamics, which could serve as a foundation for future model-based decision-making in structured domains.

I am proposing an approach that encodes graphs in latent space using a graph neural network and learns a transition model in this space. Specifically, I will compare:

a. A neural transition model (MLP)
b. A structured linear transition model (Koopman-style operator)

Predicted latent spaces are decoded back to graph structures, enabling analysis of graph evolution. A key challenge here is the mismatch between continuous latent representation and discrete graph validity. To address this, the decoder needs to be probabilistic, predicting edge existence and node attributes as distributions with constrained inference.

As the first step, I will evaluate one-step and multi-step graph prediction, and compare the Koopman-style linear dynamics model with a neural transition model, using next-step prediction accuracy, multi-step rollout stability, and generalization to unseen graph structures. The pipeline is structured as follows: Graph -> GNN encoder -> Latent representation -> Transition model -> Decoder -> Next graph.

This study aims to answer the following questions:

a. Can structured linear dynamics effectively capture graph evolution?
b. Under what conditions – graph size, evolution complexity, or rollout horizon – does the stability advantage of linear dynamics emerge, if at all?
c. Do they provide more stable long-term predictions?

d. What are the limitations of linear dynamics in discrete graph settings?

Initial experiments will use synthetic graph evolution datasets, followed by molecular graph datasets such as ZINC.

If a structured linear transition model can reliably predict graph evolution in latent space, it opens a path toward model-based planning in structured environments, where an agent can simulate future graph states before committing to an action, rather than purely relying on environment interactions. More broadly, establishing whether Koopman-style operators transfer meaningfully to discrete structured domains would contribute an empirical data point to the growing literature on data-driven dynamical systems beyond physics.